Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/15/2013
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stefan.judex@sunysb.edu	**Fax:**	FY 631-632-8577
PI Organization Type:	UNIVERSITY	**Phone:**	631-632-1549
Organization Name:	SUNY Stony Brook		
PI Address 1:	Dept. of Biomedical Engineering		
PI Address 2:	Bioengineering Building 213		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	09/30/2012	**End Date:**	12/31/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Maher, Jacilyn	**Contact Phone:**	
Contact Email:	jacilyn.maher56@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubin, Clinton (SUNY-Stony Brook)		
Grant/Contract No.:	NNX12AH32G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Editor's Note (12/17/2012): NOTE THIS IS A CONTINUATION OF FUNDING FOR NNX08BA35G WITH THE SAME PRINCIPAL INVESTIGATOR. See NNX08BA35G "Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure" for previous reporting and publications. As space missions become longer, it is apparent that long-term risks of multiple, or even single, exposures to microgravity on crew musculoskeletal health need to be defined and addressed. Even though the magnitude of the long-term deterioration in morphology and strength is largely unknown, it is clear that such risks will be heavily influenced by the genetic make-up of an individual. We have recently identified and characterized a genetically diverse mouse population that, similar to astronauts, includes a diverse range of responses to hindlimb unloading and reambulation. Using this population, here, we will address the consequences of multiple exposures to simulated weightlessness on long-term quantity and quality of trabecular and cortical bone as well as muscle, and investigate how		

	recovery periods between periods of weightlessness can be optimized. Changes in tissue morphology, density, and mechanical properties will be assessed longitudinally in each genetically distinct mouse by in vivo microCT and finite-element modeling, allowing for the identification of skeletal baseline indices that modulate changes in tissue morphology and strength during altered levels of loading. Detailed mechanical tests will be performed at the endpoints to assess the long-term quality of the tissue. Together, the proposed studies provide a unique opportunity to establish the range of responses of genetically diverse individuals to multiple unloading and reambulation phases. Optimizing the length and distribution of recovery periods between disuse exposures may ultimately help to maintain long-term musculoskeletal morphology and strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Extended periods of unloading are not confined to space flight but also occur during extended bedrest or paralysis on Earth. Therefore, data from our experiments may help understanding the basis by which the musculoskeleton responds to repeated unloading events and eventually lead to more efficacious prophylactics and treatments.
Task Progress:	This 3-month grant served as a supplement to grant NNX08BA35G with the same title to enable the ex-vivo microCT scanning of bones previously scanned by in-vivo microCT. Using techniques described previously, we subjected femurs from the second specific aim to high-resolution microCT scanning with the goal to more accurately describe differences in bone’s micro-architecture between experimental and control groups. The mechanical data will be published in the near future. Again, we would like to thank NASA for providing a 2-month graduate student stipend to accomplish these important scans. [Editor's note: See NNX08BA35G "Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure" for previous reporting and publications.]
Bibliography Type:	Description: (Last Updated: 01/16/2018)